

Mapping Vote Swing Between Elections

The election wind map, two pairs of elections, and what an arrow has to promise

84-355 Democracy's Data

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A red-and-blue map of who won tells you where the votes are. It cannot tell you what *changed*, because a county that went from 70% Republican to 60% Republican is still red, and it just moved ten points toward the Democrats.

A wind map fixes that by drawing **change** instead of **level**. Every place gets an arrow: which way it moved between two elections, and how far. Together they look like a weather chart, and they read like one — you can see a front crossing the country. It is the most legible picture of an election anyone has put on a page, and **every arrow in it makes a promise that is easy to break**.

The promise is precise. An arrow makes three claims at once. *This is one place. I observed it twice. Here is the difference.*

Now suppose the place was not the same place both times. Or suppose the two observations came from different sources that counted differently. The arrow still gets drawn. It still points somewhere. It still looks like evidence.

"This county swung twelve points" is the sentence American election coverage is built out of. It decides which places get called bellwethers, which demographic story gets written, which doors a campaign knocks on next time. This chapter draws two of these maps of the same country, four years apart, and the second one is there because a single wind map invites a conclusion that two of them will not support.

01 Three files, one of which says it is not authoritative

An arrow promises one place, observed twice. Before drawing three thousand of them it is worth asking who did the observing, because the answer is not the one the picture implies.

There is no federal source. The Federal Election Commission publishes official returns by state and by congressional district and **not by county**. County returns are produced by 51 separate state election offices in 51 formats, and no agency assembles them into anything. So every national county map anyone has ever published rests on somebody's compilation, and this one is no exception.

The national arrows on the map below come from a volunteer repository. Its own README explains where the numbers came from. The 2024 figures were scraped off a television network's results pages. The 2020 figures came from a network, a political news site and a newspaper. The README also says the numbers **"are not authoritative."**

That sentence is the reason to use it rather than a reason not to. A file that admits what it is makes a better teaching object than one that hides it. The audit in the middle of this chapter would be impossible against a source with nothing to admit.

The Georgia half of the chapter is deliberately the other kind. Both ends come from the Secretary of State's own returns, the same primary record the precinct chapter reads. And the unit has not changed since 1945.

Two maps, one encoding, two sources of very different standing. That is the comparison this whole chapter is built out of.

There is a cost buried in the word *scraped*, and it is permanent. A scrape is a reading of a web page at a moment. The network results pages those 2020 county numbers came off were overwritten in 2022, and again in 2024. The thing the file was made from is gone. Nobody can check the file against its own source, ever.

Whatever a script misread on an election night in November 2020 is now simply part of the record. It sits in a numeric column looking exactly like a count. **That is why the audit in this chapter runs sideways rather than upstream.** It compares one vintage of the file against another, and the whole national file against Georgia's official returns. Upstream is where you would normally check. Upstream no longer exists.

The county is the right unit for a picture of movement mostly by elimination. States are too coarse to show a front crossing anything, precincts are not published nationally at all, and the county is the only level in between with something like national coverage across elections. The arrows have to start somewhere. They start at the Census Bureau's population-weighted centers of population: 3,221 of them, fetched on 2026-08-10. These are built from the whole census rather than from a sample.

They are also 2020 vintage. That fact comes back at the end of the chapter.

Three vintages are read here, not two, and each was fetched and audited on its own. **The problems in this kind of file belong to a vintage rather than to a source.** The 2016 file makes that case before a single county is compared.

It arrives in an older layout, and its key column `combined_fips` is stored as an **integer** rather than as text. So Alabama's 01001 arrives as 1001. Every state whose code is below 10 has quietly lost its leading zero: Alabama, Alaska, Arizona, Arkansas, California, Colorado and Connecticut. Read that column as a number, join on it, and those states vanish from the map without any error being raised anywhere. The key is padded back to five characters and then checked, and every county identifier here is read and stored as text.

Cleaning here is therefore less about repairing values than about refusing to let a join decide anything on its own. Nothing was imputed (nothing was filled in by guessing it from something else) and no vote total was adjusted. Where two vintages disagree about what a unit *is*, the disagreement gets resolved on purpose, and the resolution is written into a table further down.

A join that returns a plausible number of rows is not evidence that it matched the right ones. That is what the next few pages are about.

Every web address the build scripts touch is recorded, with the date it was requested and the answer it gave back. Every number quoted in this chapter is read from `facts.csv`, which those scripts wrote.

There is one exception. In the section that puts this file against the fifty-one state canvasses, the numbers are computed on the page from the official returns themselves. Either way, no number in this chapter is typed in.

02 Two ways to lose a leading zero

Three vintages of the same compilation, and they do not agree about what a county identifier is. Here is the top of the 2020 file as it arrives:

Column as it arrives	What it holds	Row 1	Row 2
state_name	the state, spelled out	Alabama	Alabama
county_fips	county identifier — text on disk, five characters	01001	01003
county_name	the county's name	Autauga County	Baldwin County
votes_gop	votes for the Republican	19838	83544
votes_dem	votes for the Democrat	7503	24578
total_votes	all votes cast	27770	109679
diff	the two counts subtracted	12335	58966
per_gop	the Republican share	0.714368023046453	0.761713728243328
per_dem	the Democratic share	0.27018365142239825	0.22409029987509005
per_point_diff	the two shares subtracted	0.4441843716240547	0.5376234283682382

county_fips reads 01001. **On disk that is correct.** It is text, the leading zero is there, and Autauga County's identifier is intact.

What destroys it is the *read*. Ask for that column without saying it is text, and 01001 becomes the number 1001. That is not a county code. It matches no center point, no outline, and no other file.

Every state whose code is below 10 drops out of the join at once: Alabama through Connecticut, and California at 06. The join does not complain. A join that matches nothing just returns fewer rows.

That is why every identifier here is read and stored as text, and why the build checks the padding afterwards rather than trusting it.

Now the 2016 file from the same repository:

Column as it arrives	What it holds	Row 1	Row 2	Row 3
(unnamed)	an unnamed row counter the compilation exported by accident	0	1	2
votes_dem	votes for the Democrat	93003.0	93003.0	93003.0
votes_gop	votes for the Republican	130413.0	130413.0	130413.0
total_votes	all votes cast	246588.0	246588.0	246588.0
per_dem	the Democratic share	0.37715947248	0.37715947248	0.37715947248
per_gop	the Republican share	0.528870018000	0.528870018000	0.528870018000
diff	the two counts subtracted	37,410	37,410	37,410
per_point_diff	the two shares subtracted	15.17%	15.17%	15.17%
state_abbr	the state's two-letter code	AK	AK	AK
county_name	the county's name	Alaska	Alaska	Alaska

Column as it arrives	What it holds	Row 1	Row 2	Row 3
combined_fips	the same identifier under a different name, and unpadded	2013	2016	2020

Same compiler, same country, four years earlier, and almost nothing is the same. The key is called `combined_fips` rather than `county_fips`, it sits last rather than second, and **the leading zero is not there to lose**. 2013 is Alaska; Autauga County in this vintage is 1001. Careful reading cannot recover what was never written, so the build pads the column back to five digits and then checks that every padded code is a code the other vintages know. **This is the opposite failure from the one above, and it needs the opposite fix:** the 2020 file must not be converted, and the 2016 file must be. A build that applied one rule to both would break exactly half of its data.

Four more differences are visible in those eleven lines, and each one would have to be handled before the two files could sit in the same table. The first column of the 2016 file has no name at all — it is a row number that a naive read turns into a variable. `votes_dem` arrives as 93003.0, a whole number of ballots written as a decimal. `diff` is "37,410", a quoted string with a thousands separator inside a numeric column. And `per_point_diff` is 15.17%, a percentage carrying its own percent sign, where the 2020 file gives the same quantity as a bare proportion.

The last thing in that block is not a formatting difference and matters more than any of them. Look at the vote totals down the three Alaska rows: 3,152 rows in the 2020 file, and in 2016 Alaska's forty pseudo-counties **all carry the same numbers** — the same 93,003 and 130,413, repeated. Alaska did not report by borough, so the compiler wrote the statewide total into every row. Any county-level statistic computed over that file counts Alaska forty times, and nothing in the file says so. The arrows in this chapter's 2016 map are drawn only where a unit survived the audit two sections down, which is the whole reason that audit exists.

03 What an arrow claims

The encoding is fixed once, here, and **every map in this chapter uses the same one**. There is a single key, and both of its channels are measurements.

Direction. Straight up is no change. The arrow rotates clockwise for a shift toward the Republican candidate and counter-clockwise for a shift toward the Democrat, at 8 degrees per point. A shift of 11 points lies flat, and nothing rotates past flat — an arrow that turned further would point *down*, which would read as a third direction the data does not have. 2.1% of national counties reach that cap; no Georgia county does.

Length. How many votes moved, not how many points. Angle already carries the rate, so spending length on it too would say the same thing twice and leave the size of the place unsaid. Length carries the quantity instead: the swing multiplied by the county's two-party vote, which is the net number of ballots that changed hands. Ten points in a county of nine hundred and ten points in a county of nine hundred thousand are the same arrow under a per-point rule, and they are not the same event.

Color carries the party a second time, red for Republican and blue for Democratic, so direction survives an arrow too short to have a readable angle.

Nothing else. Every arrow is drawn at the same width. Thickness used to carry votes, badly: it is a weak channel beside position and length, and now that length carries votes properly there is nothing left for it to say.

Two choices inside that rule are worth defending, because both are decisions rather than facts about elections.

Why 8 degrees and not 5. A gentler rotation is fine on a state-sized frame and too timid on a national one. The middle county in the recent pair moved 3.1 points. At 5 degrees per point, that arrow sits 16 degrees off vertical. A county that moved the same distance the other way sits 16 degrees off on the other side. That is 32 degrees of separation, repeated across 3,100 arrows a few pixels long.

The map then reads as a field of vertical lines, and *which party gained where* becomes invisible. At 8 degrees the median lean is 24 degrees and the two parties separate by 49. Pushing further costs more than it buys. At 10 degrees per point the cap starts eating real counties, including both of the Georgia extremes this chapter names.

Why length is a square root with a floor and a ceiling. Net votes moved runs from nothing to 400,096 in Los Angeles County, 1,117 times the median county. Drawn in straight proportion the median arrow would be a fiftieth of a pixel. So length is the square root of net votes, reaching full extension at 25,000 and never falling below 15% of full.

The floor keeps a small county visible as an arrow rather than a speck. The ceiling stops one county setting the scale for three thousand. 1.9% of counties reach it.

Both are fixed constants printed in the legend, not quantiles of whatever data is in front of them.

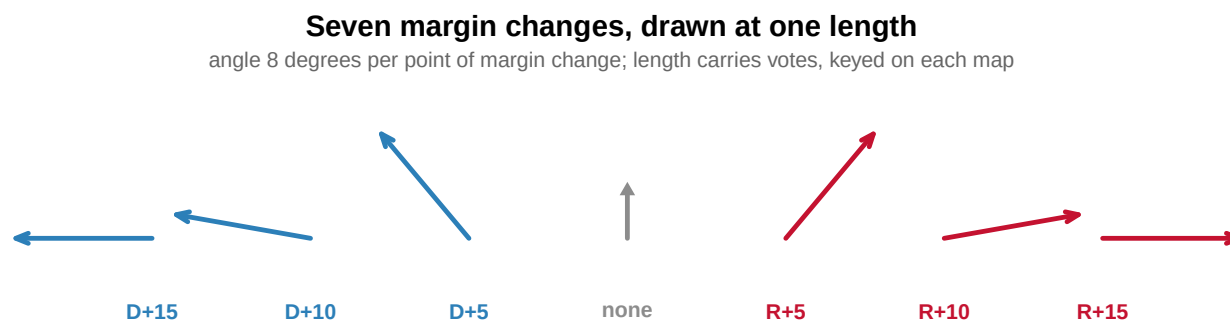


Figure 1. One arrow, seven swings. Every arrow in this chapter is one of these. The angle is 8 degrees per point, and the length is proportional to the swing. Both are real measurements. A reader with a protractor and a ruler can recover the number from the picture. Only the kilometres-per-point changes between maps, and each legend prints its own.

04 The national wind map, 2020 to 2024

Here are 3,100 counties. Each arrow starts at that county's **population-weighted centroid** — where its people actually are, not the middle of its polygon, which for a western county can be a hundred kilometers of empty desert.

How every county moved, 2020 to 2024

two-party margin; leaning right = Republican gain, left = Democratic gain

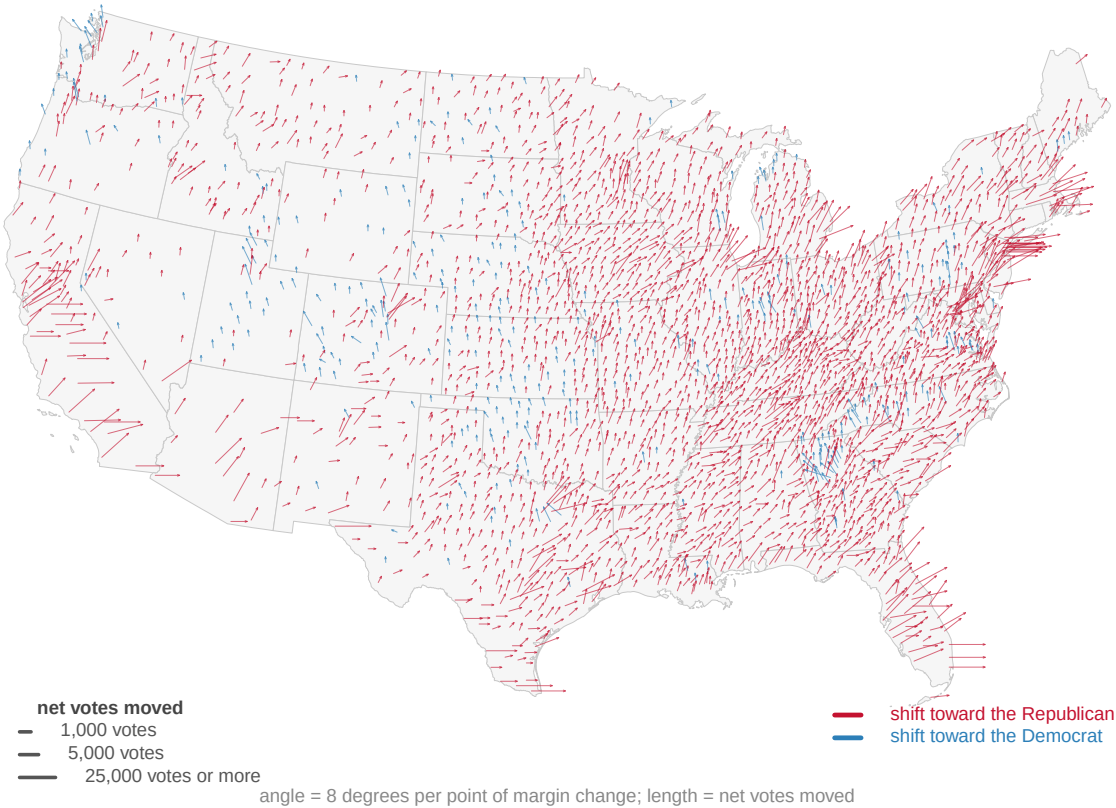


Figure 2. How every county moved between 2020 and 2024. One arrow per county. Each starts at the county's population-weighted centre — where its people actually are, rather than the middle of its shape, which for a western county can be a hundred kilometres of empty desert. Arrows lean right where the Republican two-party margin grew and left where it shrank, 8 degrees per point, flat at 11; length is the size of the net votes it moved. 88.5% of the 3,100 arrows lean right, and the field is close to uniform: this is what a national swing looks like when it is national.

The picture is unambiguous. **88.5% of counties moved toward the Republican candidate**, the median county by R+3.1, and the whole field leans one way. Only a small minority of arrows tilt left.

Quantity	Value
Counties drawn	3,100
Median county shift	R+3.1
Counties shifting right (%)	88.5
Largest shift right	R+28.1

Quantity	Value
Largest shift left	D+9.2
Arrows at the length cap	60

Hawaii's 4 counties sit outside the frame of a conic projection of the lower 48 and are not drawn. Alaska is missing for a different reason, and the reason is the next thing to look at.

05 Three ways a unit stops being the same unit

Quantity	Value
Rows in the 2020 file	3,152
Rows in the 2024 file	3,160
FIPS codes present in both	3,104
Present in 2020 only	48
Present in 2024 only	56

A first look says the country gained eight counties. It did not. Take the non-matching codes apart:

Where	Appears in	Units
Alaska	2020 only	40
Alaska	2024 only	40
Connecticut	2020 only	8
Connecticut	2024 only	9
District of Columbia	2024 only	7

Three different things are going on, and only one of them is a change on the ground.

Alaska is a labeling difference, not a geographic one. Alaska does not report presidential votes by borough at all. It reports them by State House District, and both files carry 40 of them. The 2020 file numbers them 02901-02940 and calls them "District 1" and so on; the 2024 file numbers them 02001-02040 and calls them "State House District 1". Same kind of unit, with different arbitrary codes chosen by whoever assembled each file.

No join will ever match them. And **matching them by district number would be wrong anyway**. Alaska redrew its House districts after the 2020 census, so district 12 is not the same ground in the two elections.

There is a nastier consequence. Three of the 2024 pseudo-codes collide with real Alaska borough codes in the Census Bureau's centroid file:

Fips	In the returns	In the centroid file
02013	State House District 13	Aleutians East
02016	State House District 16	Aleutians West
02020	State House District 20	Anchorage

A script that joins the 2024 returns to the centroid file gets three matches, no warning, and three arrows planted in the wrong part of Alaska.

The District of Columbia is a granularity difference. In 2020 it is one row. In 2024 it is eight rows, one per ward. Nothing moved; somebody chose to report it finer.

Connecticut looks like the real thing. The Census Bureau retired Connecticut's eight counties as statistical units and adopted **nine planning regions** as county equivalents, with new FIPS codes, effective for 2022 and after. The 2020 file uses the old codes. The 2024 file uses the new ones.

That is a real change in what a county *is* for statistical purposes. It takes 1,824,280 votes out of any county-level comparison, because the 2020-vintage centre file has no centre for a planning region either. It is also the one flag of the three that turns out not to belong to the state at all. Connecticut's election office went on reporting by the eight counties in both years.

"The same election, as the states published it," below, puts the certified returns beside the compilation and shows whose change this was.

06 The one that does not show up in a set difference

Alaska and Connecticut fail loudly: the codes do not match, the rows drop out, and you can count what you lost. **The District of Columbia fails silently.**

FIPS 11001 is in both files. In 2020 it means “District of Columbia”, 344,356 votes. In 2024 it means “Ward 1”, 40,463 votes.

Before you read on, predict. Those are the two rows the join will match: same FIPS code, District of Columbia against Ward 1. The join will match this row. **Will the resulting arrow be roughly right?** Most people say yes, reasoning that the District’s wards are all overwhelmingly Democratic, so whichever ward you happen to land on will have a margin close to the city’s. Commit to an answer – roughly right, or wrong – and to a direction, before you look.

The join succeeds. An arrow gets drawn. It compares a whole city to one eighth of itself.

One county, two joins, 3,104 rows either way

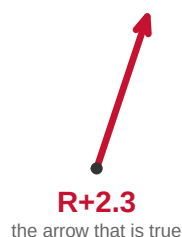
JOIN ON FIPS AND STOP

2020 District vs 2024 Ward 1



HARMONIZE FIRST

2020 District vs 2024 all wards



Both joins return the same row count. The row count cannot warn you.

Figure 3. One county, two joins, 3,104 rows either way. Both arrows are the District of Columbia between 2020 and 2024. The left one is what a join on FIPS returns if nobody reads the unit names: it compares the whole District in 2020 to Ward 1 in 2024 and reports D+0.5. The right one sums the eight 2024 wards back into one District first and reports R+2.3. Because these two arrows are a close-up rather than a national field, they use the continuous encoding of Figure 1’s top row, where the angle is itself the measurement. The two joins return an identical number of matched rows.

The arrow reverses. Joined carelessly, the District appears to have moved D+0.5, toward the Democrats. Summed properly across all eight wards, it moved R+2.3 – toward the Republicans. The error is 2.8 points, which is larger than the median county’s entire swing.

And both joins return 3,104 rows. **Row counts, matching percentages and “the join succeeded” tell you nothing about this class of error.** The only thing that catches it is reading what the units are called.

07 What the harmonization actually did

State	Problem	What we did
Alaska	Same units, different pseudo-codes, and redrawn between elections	Dropped all 40 units
District of Columbia	One row in 2020, eight in 2024	Summed the 8 wards back to one District
Connecticut	Eight counties replaced by nine planning regions	Dropped, losing 1,824,280 votes
Everywhere else	Codes agree and mean the same place	Kept

Two of those three repairs are deletions. **A wind map cannot draw an arrow for a place that did not survive**, and pretending otherwise is worse than leaving a hole.

08 The same election, as the states published it

The audit above can say **which** units broke. It cannot say **why**, and until recently nothing here could, because the compilation had nothing to be checked against. That is no longer true.

The county-returns chapter holds these two elections as the fifty-one chief election offices published them. One file per jurisdiction per year, taken from the Secretary of State or the state board of elections. Beside each sits the web address, the format it arrived in, and the state's own word on whether it was certified.

One rule there decides most of what follows: **a unit with no Census code gets an empty identifier, never a guessed one**.

The compilation is not being replaced, and no arrow in this chapter is drawn from these files. It is still the cautionary map and the audit is still the point of it. What the states' own returns add is the thing the audit was missing. Before, this chapter could say that its national file declares itself not authoritative; it could not say by how much, or whose decision each broken unit was. Now it can, and the answer is different for all three.

Unit	In the compilation	In the state's own returns	Whose change was it
Alaska	40 pseudo-coded House districts each year, renumbered between them	41 rows each year: 40 State House districts and one federal overseas district	Alaska's
Connecticut	8 counties in 2020, 9 planning regions in 2024	8 counties in 2020 and 8 in 2024, the same eight	The compiler's
District of Columbia	1 citywide row in 2020, 8 wards in 2024, all eight given codes	8 wards in 2020 and 8 in 2024, none of them with a FIPS code	The compiler's

Connecticut: the compiler changed the geography

The audit above files Connecticut as the one change of the three that is real. On the Census Bureau's side of the ledger, it is. The Bureau did retire the eight counties as statistical units, and did adopt 9 planning regions as county equivalents for 2022 and after.

Connecticut's election office did not. The Secretary of the State's certified Statement of Vote summarizes the presidential result "by County" in 8 rows in 2020 and the same 8 rows in 2024. The phrase *planning region* appears in neither document.

So the geography change that costs this chapter's later map 1,824,280 votes and 8 units was made in the compilation, not in Connecticut. And it is visible in the arithmetic. The compilation's nine 2024 regions sum to 992,053 Democratic and 736,918 Republican votes; the state's eight counties sum to 992,197 and 737,024, a difference of 144 and 106. That difference is not noise. Connecticut publishes a second, town-level export that disagrees with its own county table by exactly that much. And towns are the only unit planning regions can be built out of.

Which is to say: the compiler did not receive nine regions from anybody. They assembled them.

Here is what the audit deleted, computed from the returns Connecticut certified:

County	2020 margin	2024 margin	Swing	Two-party votes, 2024
Fairfield	D+27.5	D+19.9	R+7.6	445,282
Windham	R+4.4	R+10.4	R+6.1	55,984
New Haven	D+17.6	D+12.2	R+5.5	390,452
Hartford	D+28.1	D+22.9	R+5.2	421,938
Tolland	D+11.8	D+8.2	R+3.6	80,185
New London	D+16.4	D+12.8	R+3.5	135,161
Middlesex	D+16.6	D+13.1	R+3.5	95,827
Litchfield	R+5.1	R+8.2	R+3.0	104,392

Eight arrows, every one of them leaning the same way as the national field, carrying 1,729,221 two-party votes. Connecticut as a whole moved R+5.6, against a national R+6.0.

The state that the map reports as unmappable is in fact one of the better-documented states on it – certified, published by county, on the same eight units in both elections. The hole in New England is an artifact of the file, and this table is the size of it.

The District of Columbia: the compiler changed the grain

The audit calls the District "a granularity difference – nothing moved; somebody chose to report it finer." That is right, and the somebody can now be named. The DC Board of Elections' certified results carry a ward column in both years: 8 wards in 2020, 8 in 2024. The compilation reported the city as one row in 2020 and eight in 2024. Neither grain came from the District changing anything.

Two things follow, and they point in opposite directions.

The first is reassuring. Both files agree on every District total in both years – the compilation's citywide 2020 figure and its eight 2024 wards are the certified numbers. Sum the

wards the way this chapter does and the District moved R+2.3. That is the same figure the right-hand arrow of Figure 3 reports, and necessarily so: the numbers being summed are the District's own. **The arrow that Figure 3 calls "the arrow that is true" is the arrow the District of Columbia certified.**

The second is not. Every one of those wards is 11001 through 11008 in the compilation, and none of those codes exists. In the official file the same 16 ward rows carry 16 empty identifiers between them. The District has exactly one county-equivalent code, and it means the whole city.

The careless join that reverses the District's arrow is not possible against a file that refuses to invent an identifier. It does not quietly match Ward 1 to the city. It fails at the first row and says so.

The defect Figure 3 is built around was never in the count. It is in the key.

Alaska: the state really does report that way

Alaska is the opposite case, and the contrast is the lesson. There is no certified Alaska return by borough, in either year, because Alaska does not produce one. Its Division of Elections certifies by State House District, and the official file therefore has 41 rows in each year: 40 districts plus "District 99 - Federal Overseas Absentee". **The flag the audit raises against Alaska is a real feature of Alaska's reporting.** The flag it raises against Connecticut was an artifact of the compilation. The same audit, the same table, two completely different causes – which is exactly why an audit that only counts broken units is not finished.

Going to the source makes Alaska *more* dangerous rather than less, and this is worth sitting with. The compilation gave Alaska's districts one set of pseudo-codes in 2020 and a different set in 2024, so a join across the two matches nothing and the problem announces itself. The official file gives both years the same stand-in code: the district number with 02 in front. So a 2020-to-2024 join on it matches 41 of 41 rows and returns arrows for all of them. Every one would be wrong. Alaska redrew its House districts after the 2020 census, so district 12 is not the same ground in the two elections.

Three of those stand-in codes also collide with real borough codes in the centre file, exactly as the compilation's 2024 codes do: 02013, 02016, 02020. **A file being official does not make a join safe.** It makes the numbers right and leaves the key exactly as dangerous as it was.

Two arrows nobody flagged

The audit's three units were the ones a set difference could find. The states' returns turn up two more that no set difference ever would, because both involve codes that match perfectly.

Cumberland County, Maine, and an arrow that reverses. Maine publishes its military and overseas ballots as a single statewide row that belongs to no county. It held 4,599 Democratic and 838 Republican votes in 2020, and 5,310 and 986 in 2024. In 2020 the compilation left them out, and its sixteen Maine counties match the state's canvass exactly.

In 2024 it put all of them into Cumberland County. The compilation's Cumberland is the certified Cumberland plus exactly 5,310 Democratic and 986 Republican votes – which is

the statewide overseas row, to the ballot. Same file, same state, two vintages, two different treatments. Nothing anywhere says so.

The arrow the map draws for Cumberland is D+0.5. The arrow Maine's returns support is R+0.5. **It is not a small error; it is the other direction.**

Jackson County, Missouri, and an arrow for a unit that is not on the map. Missouri's canvass reports Kansas City as its own returning jurisdiction, separate from the four counties it straddles. The compilation folds the whole city into Jackson County. That is not a wrong number. Add Kansas City back to the official Jackson row and you recover the compilation's figures to the vote. But it is a different place.

Jackson County as Missouri returns it swung R+0.1. Jackson County with Kansas City folded in swung R+2.8, and that is the arrow on the map. **Two defensible units, 2.8 points apart, and the map does not say which one its arrow is for.**

Neither does the official file, and it cannot. Parts of Kansas City lie in three other counties, and Missouri publishes no way to split it.

How far apart the two files actually are

The last thing the ruler is good for is the size of the doubt, which this chapter has so far been able to describe only as a disclaimer.

Quantity	2020	2024
County rows checkable against the state's canvass	3,106	3,138
Rows with no official counterpart	46	22
Rows differing on a major-party count	138	47
Median state error, two-party margin	0.000 pts	0.000 pts
Worst state, two-party margin	Maine 0.41 pts	Alaska 0.10 pts

At the level the national maps are usually read, a field of arrows and a country leaning one way, the compilation holds up.

44 of the 51 jurisdictions reproduce the certified 2024 two-party margin to the last decimal the arithmetic has, and 37 do for 2020. Only 1 state in 2024, and 4 in 2020, are off by more than a tenth of a point. Nationally, the Associated Press's certified state totals give 75,019,605 Democratic votes. The compilation gives 75,006,739 and the states' own files give 75,018,411. That leaves the compilation 12,866 low and the canvasses 1,194.

At the level of a sentence about one county, it does not hold up. The 2020 file is the weaker of the two: 138 counties disagree on a major-party count, against 47 four years later.

The 2020 disagreements are not scattered. 102 of Kansas's 105 counties differ, 101 of them low, and the state is short 24,960 two-party votes in the compilation. That is the signature this chapter already knows: an election-night reading, taken before the slow ballots were counted, in a state that counts slowly. The 2016 file was replaced for exactly this. The same fingerprint is on the 2020 file that every arrow in Figure 2 is drawn from. It is smaller, one state rather than a coast, and it is the same thing.

09 The same map, one election earlier

Figure 2 is a map of a country moving one way, and it is very hard to look at it without concluding something. The obvious conclusion is that a country whose counties nearly all move toward one candidate is a country that elects him. Which it was. So the map appears to have explained the election.

The way to test a picture like that is to draw the same picture of an election that came out the other way: **2016 to 2020**, an election in which Donald Trump lost. Same encoding, same counties, same country.

Not, however, the same source, and the reason is the most useful thing in this section. The obvious move was to take 2016 from the repository the other two years come from, which does go back to 2008. That file turned out to be an **election-night scrape**, read off news results pages before the states had finished counting. So the ballots counted later are simply absent.

A shortfall like that is dangerous exactly because it is not random. Missing votes sit where counting is slowest, and counting is slowest where people vote by mail. In that file Arizona is holding 79% of its final total, California 84%, Utah 87%, New York 91%; 11 states are below 95%. Texas and Florida, which finish on the night, sit at 99% and 100%. So the error is a *map* of how states administer elections, and a national picture built on it would have drawn the West wrong while looking exactly as confident as this one does.

So 2016 here comes instead from the **MIT Election Data and Science Lab**, whose county file is built from official state records rather than from news pages. A separate script scores both candidates against a third file that neither is built from: the state-level returns already sitting in this book for another chapter. It writes the result to its own table.

On the two-party margin the arrows actually draw, the typical state error falls from 0.149 points to 0.004. That is the same accuracy as the 2020 and 2024 files. The worst single state, CO, was off by 2.19 points.

6,578,910 votes come back. The national aggregate swing moves from D+3.5 to D+2.3, against D+2.3 computed from the FEC's own national totals.

A national error rate hides the thing worth seeing, so here is the correction itself. Every county whose arrow moved is a dot, placed where the county is.

Where the old 2016 file was wrong

each dot is a county whose 2016-to-2020 swing moved when the scrape was replaced

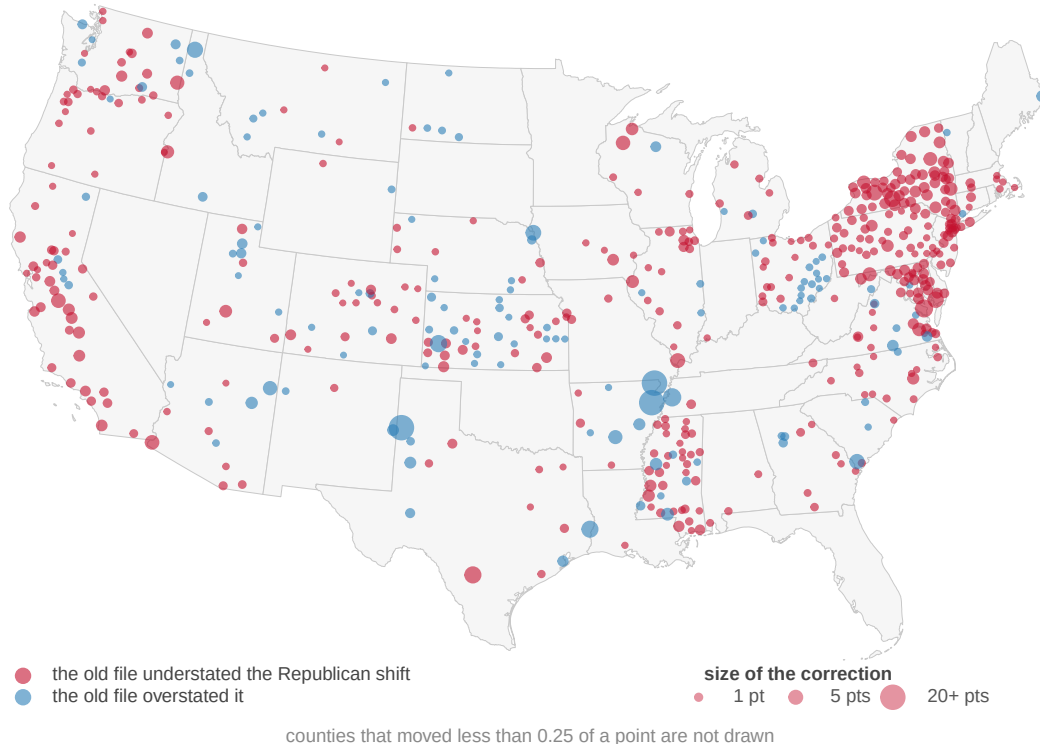


Figure 4. The correction, mapped. 532 counties had their 2016-to-2020 swing move by more than 0.25 of a point when the election-night file was replaced with official returns. 188 moved by more than a full point, and 39 changed direction outright.

Two dense clusters show up: one through New York, New Jersey, Pennsylvania and New England, and one down the Pacific coast into Arizona. That is the late-counted mail vote arriving. Across almost all of it the old file **understated** how Republican the shift had been, because the ballots it never saw leaned Democratic.

But look at Arkansas, Mississippi and Missouri. There the dots run the other way. They include the largest single correction of all: Poinsett County, drawn by the scrape at R+35.9 and by the official returns at R+12.1.

Those states do not count slowly. Late counting explains the shape of this map. It does not explain all of it. A file is allowed to be wrong in more than one way at a time.

This has a cost, and the cost belongs in the open. The two ends of this pair now come from sources of different standing: 2016 official, 2020 a press compilation. A swing is a difference, and a difference between two differently made numbers inherits both of their errors. The trade is still the right one. A two-point error in a state's margin is a wrong arrow, while a mixed pair of sources is only a caveat about right ones.

But it does mean this map and the 2020-to-2024 map are not quite the same kind of object, and neither one says so on its face.

The join has to be audited again, because a third vintage brings a third set of unit problems, and they are not the same ones:

Vintage problem	In 2016 to 2020	In 2020 to 2024
Alaska	Dropped. Reported as 40 State House districts, which have no county to join to	Dropped. Two sets of pseudo-codes for 40 redrawn House districts
District of Columbia	One row in both years; folded anyway, and unchanged by it	One row in 2020, 8 wards in 2024
Connecticut	No problem: 8 counties in 2016, 8 in 2020	Dropped. Eight counties became nine planning regions, losing 1,824,280 votes
Everything else	Codes agree	Codes agree

Connecticut survives this pair. The 2016 and 2020 files both use the old eight-county codes, so the change that costs the later pair 1,824,280 votes does not arise here at all. That is a reminder that a data problem belongs to a *pair of vintages* rather than to a dataset.

What the 2016 join loses instead is Alaska, and almost nothing else. It drops 41 units and 394,546 votes, which is 0.25% of the 2020 total. Of those, 391,346 are Alaska's. 3,107 arrows survive to be drawn, against 3,100 on the later map.

How every county moved, 2016 to 2020

same encoding as Figure 2; leaning right = Republican gain, left = Democratic gain

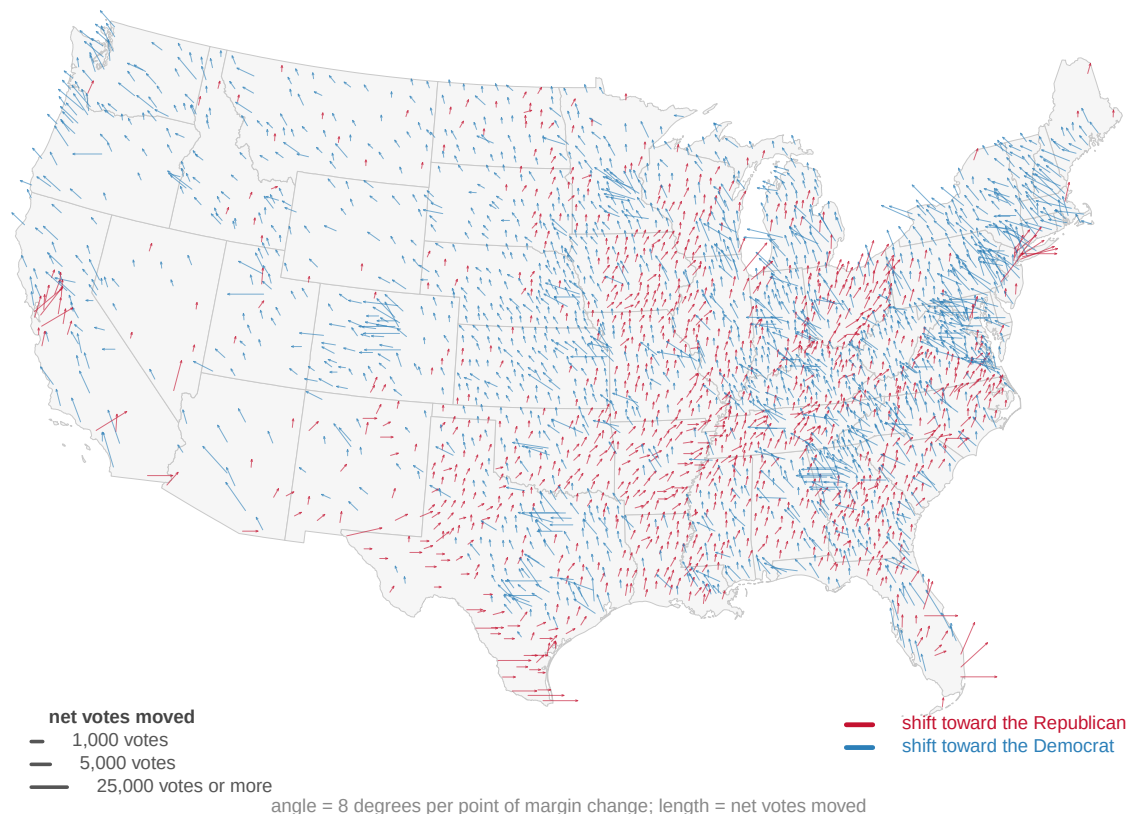


Figure 5. The same country, the same encoding, the previous pair of elections. 3,107 counties between 2016 and 2020. They are drawn at the same length-by-votes scale and the same 8-degree rotation as Figure 2, so the two pages can be laid side by side. 38.3%

of these arrows lean right, against 88.5% in Figure 2. The median county moved D+1.3, and the national two-party margin moved D+2.3. Donald Trump lost this election.

It is not the same *file*, though. The 2016 end comes from official state records rather than from the press compilation behind Figure 2, for the reason given above.

10 Which margin did this arrow draw?

Figures 2 and 4 look like readings. They are readings of a quantity that has to be defined before it can be read, and the definition is not on the page.

Margin sounds like one thing. It is at least two.

A **two-party** margin is Republican votes minus Democratic votes, divided by the sum of those two. It holds third parties constant, and asks only how the people who chose between the two major candidates divided. An **all-vote** margin divides by every ballot cast, third parties included. So it moves when a third-party vote collapses, even if not one voter switched between the majors.

Both are defensible. Both are in routine use. Every arrow in this chapter draws the first one.

Here is what the two maps say about themselves under the definition they were drawn with.

Measured over	2016 to 2020	2020 to 2024
Counties drawn	3,107	3,100
Counties moving toward the Republican	1,190	2,744
...as a share of counties	38.3%	88.5%
...as a share of the votes cast	26.5%	91.1%
Counties moving toward the Democrat	1,917	356
...as a share of counties	61.7%	11.5%
...as a share of the votes cast	73.5%	8.9%
Median county shift	D+1.3	R+3.1
National two-party margin moved	D+2.3	R+6.0
Who won	Biden	Trump

Note in passing what the last two rows do *not* show. In the earlier pair the county majority leaned toward the Democrat and the Democrat won; in the later pair it leaned toward the Republican and the Republican won. The county majority agreed with the outcome both times. That is worth knowing mostly because it is the reverse of what a picture of 3,100 equally-sized arrows is usually wheeled out to demonstrate, and because two cases is not a finding.

Now change the definition and nothing else. Third parties collapsed between 2016 and 2020, from a substantial national share to almost nothing, so the two margins have very different amounts of room to disagree in the two pairs.

Counties moving toward the Republican	2016 to 2020	2020 to 2024
Two-party margin (what every arrow above draws)	38.3% of counties, 26.5% of votes	88.5% of counties, 91.1% of votes
Margin as a share of all votes cast	51.2% of counties, 26.2% of votes	89.9% of counties, 91.9% of votes

Change the definition and the 2016-to-2020 answer moves by 12.9 points — from 38.3% of counties to 51.2%, close enough to an even split that either headline could be written. The 2020-to-2024 answer barely moves at all, from 88.5% to 89.9%. Nothing about the data moved between those two readings. Same counties, same votes, same arithmetic, one word redefined. One of the two pairs holds up under it and the other does not.

A wind map never prints which margin it drew. It shows you an arrow. The arrow in Figure 5 leans left in 61.7% of counties under one defensible definition, and in 48.8% under another equally defensible one.

That difference measures nothing that happened to a voter. It is what became of the people who voted for neither major party in 2016.

That is the asymmetry worth carrying out of this chapter. **The 2020-to-2024 headline survives the choice and the 2016-to-2020 headline does not. Nothing on the face of either map tells you which kind you are looking at.**

Both were drawn the same way, at the same scale, from counties counted the same way. A reader given only Figure 5 has no way to learn that its central number moves 12.9 points under a definition the map never mentions. A reader given only Figure 2 has no way to learn that its number is solid.

The map is equally confident either way. Confidence is the one thing an arrow cannot help expressing.

And the second row of the table is where the land-versus-population argument lives. Under the all-votes definition, the counties that moved toward the Republican in 2016-to-2020 are 51.2% of the counties and 26.2% of the votes. **A nearly even split of the map is a three-to-one split of the electorate.** Counting counties and counting voters are different operations, and only one of them elects anybody.

11 Eight years, drawn once

The two national maps cover four years each and disagree with each other. That invites an obvious question, and it has a testable answer: **what happens if you draw the whole eight years as one arrow?**

There are two plausible predictions, and they point opposite ways.

Regression to the mean. This is the tendency of an extreme reading to drift back toward the middle next time you look. The country swung left between 2016 and 2020 and right between 2020 and 2024. Suppose those were largely the same places lurching one way and then back. The two halves would partly cancel. The eight-year map would be calmer than either four-year map, and the apparent trends in both would turn out to be noise around a stable position.

Durable realignment. Now suppose the opposite. Suppose the places that moved most in the first period were the places that moved most in the second, in the same direction,

after allowing for whichever way the national tide was running. Then the halves would compound, and the eight-year map would show larger and better-organised arrows than either half.

These are distinguishable without looking at the map at all, which is the right order to do it in.

The halves accumulate; they do not cancel. Two four-year swings laid end to end with no relationship between them would give an eight-year swing with a standard deviation of 6.1 points. The observed figure is 7.3 — larger, not smaller, and larger than either half on its own. The correlation between a county's 2016-to-2020 swing and its 2020-to-2024 swing is **0.44**: positive, and not weakly.

Per county or per voter? Every number in the last paragraph counts counties. That gives Loving County, Texas, with 96 two-party votes, the same say as Los Angeles County's 3,606,971. That is a ratio of 37,573 to one. Treating it as one-to-one is not a rounding error. It is a named fallacy.

Grofman and Cervas (2024) catalogue *failing to weight units* among the arithmetic errors that recur in claims about American election results, alongside the observation that the largest 150 counties hold about half the votes cast. In this file it takes 158 of 3,099 counties to reach half the two-party vote.

So the correlation is computed both ways. The choice does not rescue the mean-reversion story. **It makes the build-up stronger.** Counting each county by its two-party vote raises the correlation from 0.44 to **0.62**. It raises the fitted slope from 0.28 to 0.49. The weighted spread of the eight-year swing is 9.1 points, against 7.2 if the halves were unrelated.

The persistence is a fact about where voters are, not a side effect of counting empty counties.

It does change one headline. 63.1% of *counties* lean right over the eight years; 56.6% of the votes do. The two numbers describe the same eight years and disagree by 6.5 points, which is the same gap section 10 measures on the map.

That positive correlation is worth reading slowly. 52.1% of counties moved toward the Democrat in the first period and toward the Republican in the second. That is a sign flip in a majority of the country.

The sign flip is the national tide reversing, and it happened to almost everyone. The correlation is about what is left after the tide. *Relative to their peers*, the counties that moved furthest toward the Republican in 2016-to-2020 are the counties that moved furthest toward the Republican again in 2020-to-2024. Only 2.0% of counties went the other way, right and then left.

So the eight-year map is not the two four-year maps averaged into calm. It is the two of them added up.

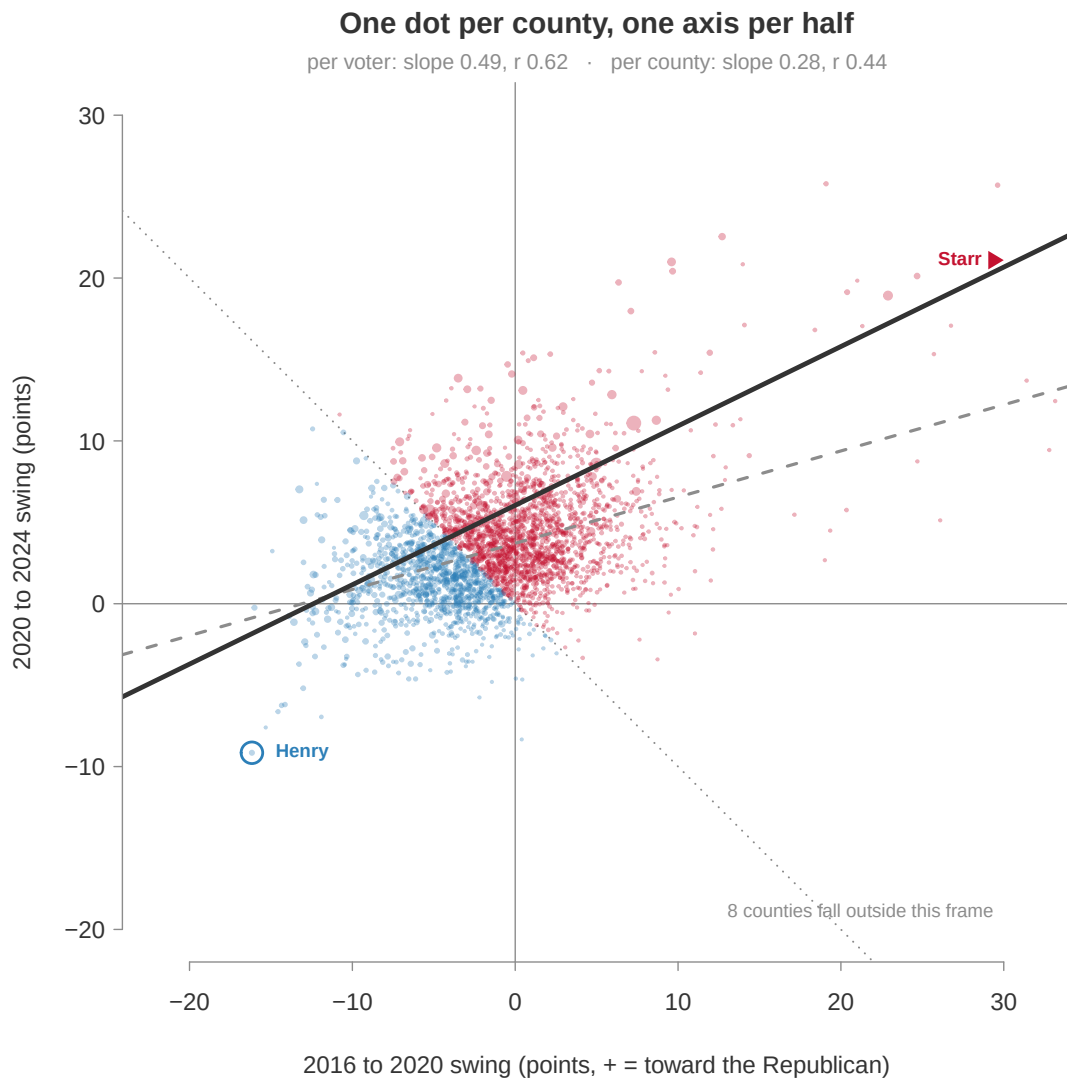


Figure 6. The two halves, plotted against each other. Each dot is one county. Its 2016-to-2020 swing runs along the horizontal axis and its 2020-to-2024 swing up the vertical. Dots are sized by votes cast and coloured by which way the county moved over the whole eight years. **If the halves cancelled, the cloud would run down the finely dotted line.** It runs up instead.

Two lines are drawn, because they answer different questions. The dashed line counts every county equally, at slope 0.28. The solid line counts each county by its two-party vote, at slope 0.49. The axes are equally scaled, so those tilts are real angles and not a choice of frame.

Two features are worth naming.

The cloud sits mostly *above* the horizontal axis and mostly *left* of the vertical one. That is the pair of national tides, D then R, and it is why a majority of counties changed sign. And it leans upward anyway. That is the part the tides do not explain: the counties furthest right in the first period are furthest right again in the second.

Starr and Henry are labelled here and are the longest arrows in either direction on the map below. Starr's first-half swing of R+56.3 runs off the right-hand edge. It is drawn as an arrowhead on the frame rather than as a dot inside it, along with the 8 other counties

the frame excludes.

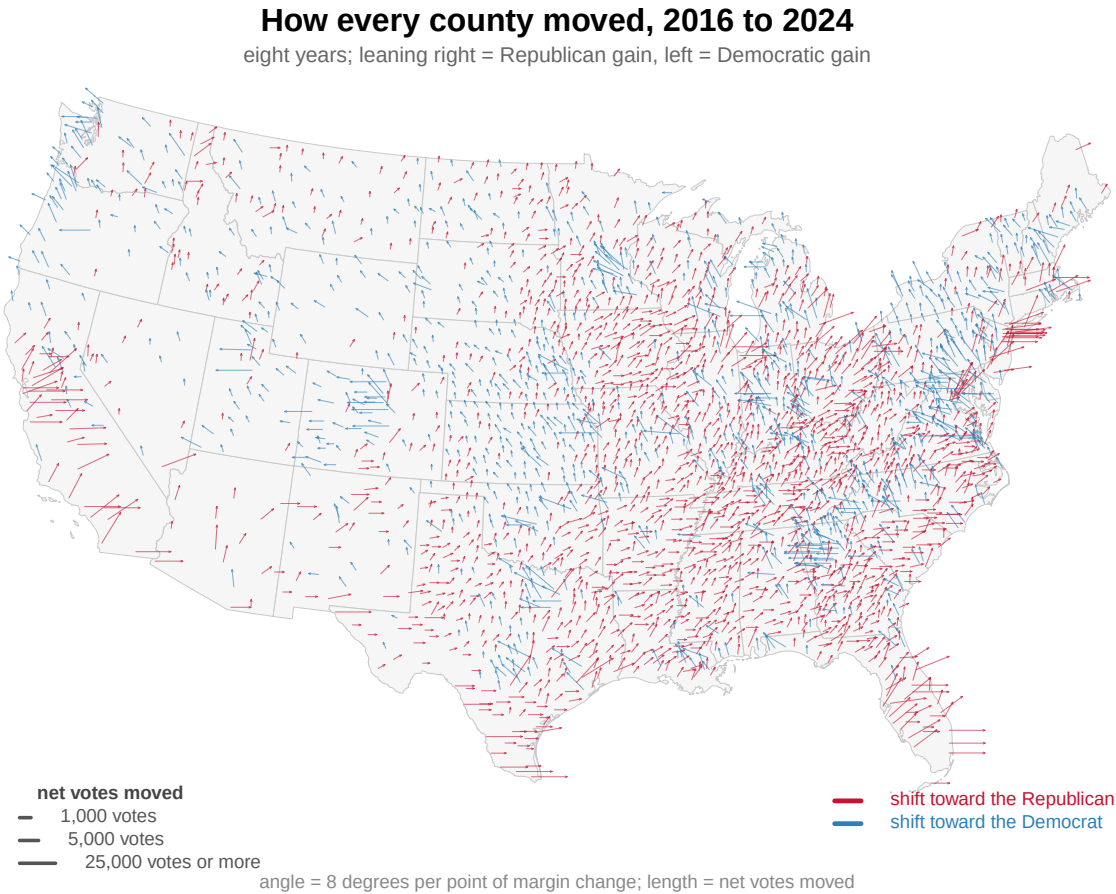


Figure 7. Eight years as one arrow, 2016 to 2024. 3,099 counties, the same encoding and the same scale as Figures 2 and 4, so all three can be laid side by side. 63.1% of counties lean right over the eight years – between the 88.5% of 2020-to-2024 and the 38.3% of 2016-to-2020, as an accumulation of the two has to be. The median county moved R+1.8 and the national two-party margin moved R+3.8, from D+2.0 to R+1.7.

Quantity	Value
Arrows drawn	3,099
Counties leaning right (%)	63.1
...as a share of the votes cast	56.6%
Median county shift	R+1.8
National two-party margin moved	R+3.8
Spread of the eight-year swing (sd)	7.3
...if the two halves were unrelated	6.1
Correlation between the two halves	0.44
Biggest shift right	Starr, Texas R+77.4
Biggest shift left	Henry, Georgia D+25.3
Arrows at the length cap	70

Where the accumulation actually is. The longest arrow on the map is Starr, Texas, which went from D+61.4 to R+16.1 — a R+77.4 shift, and the whole lower Rio Grande around it moves with it. The longest arrow the other way is Henry, Georgia, at D+25.3, in the ring of counties south-east of Atlanta. Neither of those is visible as a *front* on either four-year map, because on each of them it is half-built. Eight years is the span on which they read as places that have changed rather than places that had a year.

And the eight-year map costs a state. Connecticut's 8 counties cannot appear here at all. The 2016 and 2020 files use the eight historic county codes; the 2024 file uses the nine planning regions that replaced them. Figure 5 keeps Connecticut because both ends of that pair predate the change. Figure 2 loses it and so does Figure 7, for the same reason and at both ends. **The join is a property of the pair of vintages, not of the file** — which is the point section 4 made about Alaska, arriving a second time and costing a different state.

12 One arrow per county, or one arrow per voter?

Look again at Figure 2 and notice what it is counting. There are 3,100 arrows and every one is the same size. Los Angeles County cast 3,606,971 two-party votes; the smallest county drawn cast 96. That is a ratio of about **37,573 to one**, and on the page they are two arrows.

The map partly compensates, in that arrows are drawn thicker where more votes were cast. But thickness is a weak signal next to position and length, and there are thousands of thin arrows and only a few fat ones.

Question	Answer
What did the typical county do?	mean county shift R+3.4
What did the typical county do? (median)	median county shift R+3.1
What did the votes do?	national two-party margin moved R+6.0

The counties moved about R+3.1. The country moved R+6.0 — roughly twice as far. Both numbers are correct. They answer different questions, and the wind map answers the first one while looking like it answers the second.

This is an editorial choice rather than a technical detail. **One arrow per county gives rural America roughly ten times the visual weight it has in the electorate.** Newspapers make that choice constantly, usually without saying so. This chapter makes it too. The difference is that it is being said.

13 Georgia, from the state's own record

The national map is drawn from a compilation nobody certifies. Here is the same encoding on data that is somebody's legal record.

Both years come from the Georgia Secretary of State. The 2020 returns are the November 3 general election archive, one XML file per county. The 2024 returns are the state's own results export. Georgia has had exactly **159 counties since 1945**, so no unit changed,

no crosswalk was needed. A crosswalk translates one set of areas into another, and no arrow here can be fictional.

This is also the map that keeps the **continuous** encoding of Figure 1's top row. There are 159 arrows on a state-sized frame here, rather than 3,100 on a national one. That leaves room between them to read an angle, so the rotation goes back to being a real measurement: 8 degrees per point, recoverable from the picture with a protractor.

Georgia, county by county, 2020 to 2024

Secretary of State returns, both years; 159 counties, unchanged since 1945

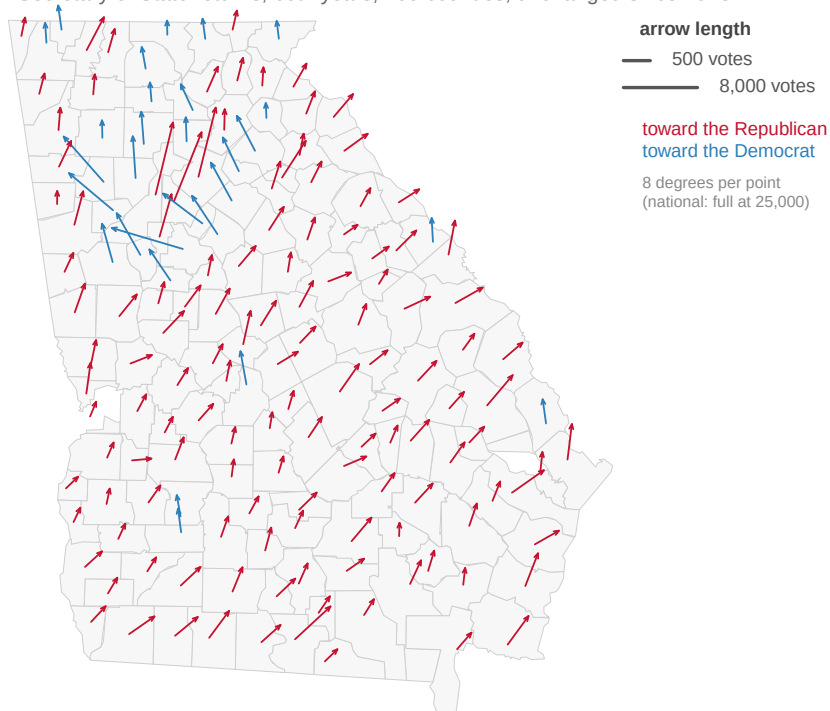


Figure 8. Georgia between 2020 and 2024, from the state's own returns. 159 counties, with Secretary of State returns on both ends. This is the only map in the chapter whose arrows can be read as angles. Rotation is 8 degrees per point, and length carries net votes moved.

81.8% of counties leaned right, the middle one by R+3.0. The statewide margin moved R+2.5, from D+0.3 to R+2.2. Nothing here needed a crosswalk, because the unit has been the same since 1945.

Quantity	Value
Counties drawn	159
Georgia margin, 2020	D+0.3
Georgia margin, 2024	R+2.2
Statewide shift	R+2.5
Median county shift	R+3.0
Counties shifting right (%)	81.8
Biggest shift right	Webster R+10.6
Biggest shift left	Henry D+9.2

A note on comparing the two maps. Angles mean the same thing everywhere. 8 degrees is one point of margin change on every figure here, so a lean can be read off any of them without converting anything. Lengths cannot be compared across maps, and that is deliberate. Georgia reaches full extension at 8,000 net votes and the national maps at 25,000. A state-sized frame and a country-sized one do not hold the same electorates. Both numbers are printed in their legends. Anything looser and a wind map becomes a picture of the size of its own frame.

14 Why not precincts? A join, measured

Georgia has 2,701 precincts. That is roughly **17 times the detail** of the county map. A companion chapter has already done the hard part: the 2020 votes have been carried onto the 2024 precinct boundaries using census blocks, so the boundary problem is solved. All that remains is attaching the 2024 returns, and the only key available is the precinct **name**. Here is what each obvious repair to that join buys:

Rung	Matched	Unmatched 2020	Pct of 2020
1. exact name, uppercased	2198	494	81.6
2. strip leading ballot-order number	2353	339	87.4
3. strip trailing (CITY) tag	2354	338	87.4
4. strip all punctuation and spaces	2446	246	90.9

Quantity	Value
2020 precincts to place	2,692
Placed after every repair	2,446
Left unplaced	246
Share unplaced (%)	9.1
Counties affected	44
Worst county	CHATHAM (Savannah)
Precincts unmatched there	86 of 86

The ladder stalls at 90.9%.

And what is left over is not scattered. It piles up in 44 counties. **CHATHAM County loses all 86 of its precincts.**

The returns identify precincts by a code like 1-05C. The boundary file identifies them by the name of the polling place. No amount of punctuation-stripping bridges that.

This is why Figure 8 is drawn at county level. A precinct map here would have a hole where Savannah is, and the hole would not be visible on the page. It would just be a part of the state with no arrows, which most readers would take to mean “nothing happened there.”

A join that loses 9% of its rows in one place is worse than a coarser unit that loses none. Resolution you cannot trust everywhere is not resolution.

15 The 2026 arrow, and the office problem

This chapter has a second half that does not exist yet. The midterm is **Tuesday 3 November 2026**, during the semester, and the plan is to draw the same map live once returns are available.

Which raises a question with no comfortable answer. **A 2024-to-2026 arrow would compare a presidential race to a midterm** – different office, different electorate, different turnout. Is that a swing?

It is not, and the size of the problem can be measured almost exactly. Take Georgia's November 2020 ballot, which carried a presidential race *and* a US Senate race. Same voters. Same day. Same piece of paper. Any disagreement between the two offices is a **pure office effect with zero turnout difference** – a floor under the error a cross-office arrow carries before it has measured anything real.

Quantity	Value
Georgia margin, president	D+0.2
Georgia margin, US Senate	R+1.8
Gap between two offices on the same ballots	2.1 points
Median precinct-level gap	1.6 points
Precincts differing by over 2 points (%)	42.0
Largest precinct gap	24.6 points
For comparison: Georgia's whole 2020 to 2024 shift	2.5 points

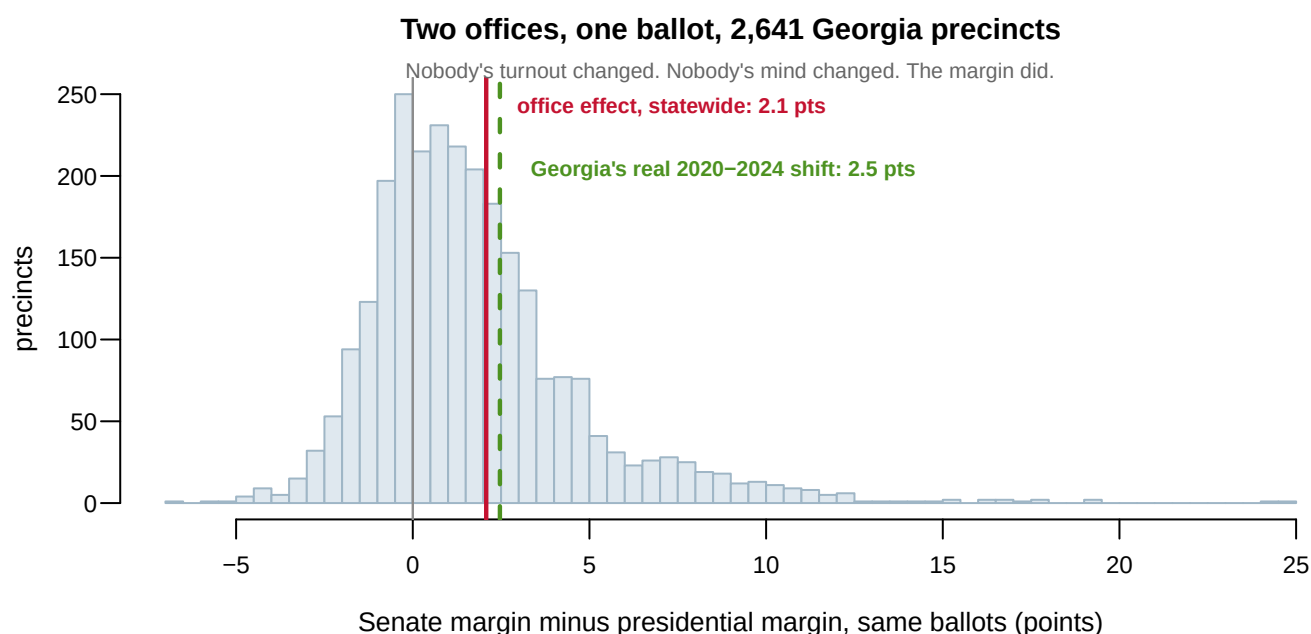


Figure 9. Two offices, one ballot, 2,641 Georgia precincts. Each precinct's US Senate margin minus its presidential margin, on the same November 2020 ballots. Nobody's turnout differs between the two numbers, because they are the same voters on the same day; every point of disagreement is a pure office effect. The red line is the statewide gap, 2.1 points. The dashed green line is the whole of Georgia's real shift between two presidential elections, 2.5 points. The distribution is wide enough that 42.0% of precincts differ by more than two points before anything real has been measured.

The office effect alone is 2.1 points – comparable to the entire real shift Georgia experienced between two presidential elections. Half of precincts differ by more than 1.6 points. And this is the *best case*: identical voters, identical day. A 2024-to-2026 comparison adds a turnout collapse of roughly a third on top of it.

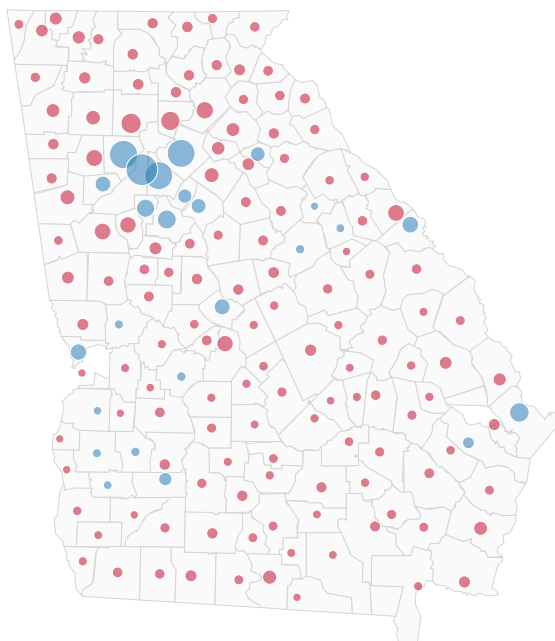
So this chapter does not draw a 2024-to-2026 arrow. The map to draw compares **Georgia's 2022 governor's race to its 2026 governor's race**: same office, both midterms, same unit, same publisher. The 2022 half is already built.

Option	Office	Electorate	Verdict
2024 president to 2026 governor	different	presidential vs midterm	carries at least 2.1 points of office effect before anything real
2022 governor to 2026 governor	same	midterm vs midterm	what this chapter uses

16 What is on the screen a month after the election

The 2026 map, waiting: 159 Georgia counties, 2022 governor

Kemp vs Abrams, R+7.6 statewide. Every dot is an arrow with only one end.



The second end arrives on 3 November 2026.

Figure 10. 159 arrows with only one end. Each dot is a Georgia county's margin in the

2022 governor's race, colored by which party carried it and sized by votes cast; statewide that race was R+7.6. There is no arrow because an arrow needs two observations and only one of them has happened. The second end arrives on 3 November 2026, and everything above about units, publishers and offices is what has to be settled before the line can be drawn.

Each dot is a county's 2022 governor margin, sized by votes cast. **These are the tails of 159 arrows whose heads do not exist yet.**

What will and will not be knowable a month after election day is worth stating before the day arrives, because the answer is not "everything":

By early December	Expect
Unofficial county returns for most states	yes, from press compilations
Georgia's own results export	likely, and it is the source this chapter uses
Certified, canvassed returns everywhere	no; certification deadlines run into December and beyond
Every race called	no; close races and recounts will still be open
Precinct-level returns	rarely this fast

The honest position is that the map drawn in class will be provisional, and that the right response is to label it so, not to wait. Every wind map published on election night has this property. Most do not say so.

17 What the argument about wind maps is really about

Wind maps are contested, and not only on aesthetic grounds.

The case for them is that they show change rather than level, which is what almost every election question is actually about, and that they do it without throwing away geography. The case against has three parts worth stating fairly.

First, they overstate the confidence of small places. An arrow computed from 96 votes is drawn with the same crispness as one computed from 3,606,971. There is no error bar on an arrow and no obvious way to put one there.

Second, direction is a metaphor that does not survive scrutiny. An arrow pointing north-east suggests movement toward somewhere. Nothing moved. Nobody walked east. The visual grammar borrows its persuasiveness from weather maps, where the arrows really do mean a thing is traveling.

Third, the margin is not the only thing that changed. A county whose margin held steady while turnout fell by a third gets a stubby arrow that says "no change." Something changed a great deal.

Fourth, and the objection this chapter is built around: the arrow is drawn from a join, and a join reports its successes, never its meanings.

Fifth, the direction of the majority is not the direction of the country, and neither is a direction that survives redefining a margin. The two national maps here are drawn with the same rule. The earlier one changes its headline by 12.9 points of counties, depending on whether third-party ballots sit in the bottom number. Neither version is wrong. Nothing on the page says which one is on the page.

18 What 6,207 arrows can testify to

The strength of this evidence is uneven by design, and the unevenness is the lesson. The Georgia map is drawn from the state's own certified record on both ends, on a unit that has not changed since 1945, so no arrow on it can be fictional. The centroids underneath every map are a complete census construction rather than a sample. And the arrow geometry is computed once and consumed identically by the interactive and the printed version of each figure, so the two renderings cannot quietly disagree.

Its limits each hold back a different sentence.

The 2020 and 2024 national returns are a press compilation that says so itself. There is still no federal publisher of county-level presidential results, so there is no authoritative national version to check them against. But there are fifty-one state canvasses, and they have now been read. 138 counties disagree with the compilation on a major-party count in 2020, and 47 in 2024, while 44 of the 51 jurisdictions reproduce the certified 2024 two-party margin exactly.

The 2016 returns are official. That sounds like a pure improvement and is not quite one. It means the 2016-to-2020 arrows measure a difference between two kinds of number, while the 2020-to-2024 arrows measure a difference between two of the same kind.

1,824,280 Connecticut votes are missing from the 2020-to-2024 map, along with all of Alaska and Hawaii. The 2016-to-2020 map keeps Connecticut and loses only 394,546 votes, 0.25% of the total, almost all of it Alaska.

Margin swing ignores turnout entirely. A county whose margin held while a third of its voters stopped voting draws a stubby arrow that says nothing changed.

60 arrows on the later map and 54 on the earlier one are at the length cap, and understate their county's movement. **The centre file is 2020 vintage**, so it puts 2024's voters where 2020's people lived. And **direction on the national maps is only a lean**. 8 degrees per point either way says which party gained and nothing about how much. That is a real loss, traded for a map anyone can read.

Three questions remain genuinely open.

The first is whether a national county-level election file should exist at all. It is not obviously a gap. The absence may be a feature of federalism, and filling it would mean somebody deciding what a county is in years when states disagree. The second is how to draw uncertainty on an arrow.

An arrow computed from 96 votes is drawn exactly as crisply as one computed from 3,606,971, and there is no obvious place to put an error bar on a direction. And whether the 2026 map should be drawn in early December at all is the third, given how much of the count will still be uncertified.

What can be said is smaller than two full-page maps suggest and more durable. A wind map encodes a swing, and on the Georgia map the number can be read back out of the angle at 8 degrees per point. An arrow promises one place, observed twice, by the same rule both times.

That promise breaks in at least three ways, and only two of them are loud: a unit that was recoded, a unit reported at a different grain, and a unit that genuinely stopped existing. The dangerous one is the quiet one, where the join returned 3,104 successful matches either way and the District of Columbia's arrow pointed the wrong direction.

One arrow per county is an editorial position, not a technical detail. The counties moved R+3.1 and the country moved R+6.0. And the three national maps together say things none of them says alone.

A page covered in arrows leaning the same way is a picture of land. Its central number may or may not survive a change in the definition of *margin*. The later pair's 88.5% moves 1.4 points under that change; the earlier pair's moves 12.9. Both maps look equally sure of themselves.

Run the span out to eight years and the two four-year swings turn out to compound rather than cancel. The correlation is 0.44, and the spread is 7.3 points where 6.1 would mean the halves were unrelated.

So the fronts that look like weather on either four-year map are mostly the same places, moving the same way, twice.

19 Sources

Data

- Georgia Secretary of State, *Election Results*. November 3 2020 general election archive; November 2024 and November 2022 results exports. <https://sos.ga.gov/page/historical-elections-results> and <https://results.sos.ga.gov/>
- U.S. Census Bureau, *Centers of Population by County: 2020*. https://www2.census.gov/geo/docs/reference/cenpop2020/county/CenPop2020_Mean_CO.txt
- County presidential returns, 2020 and 2024 — Tony McGovern, *US County Level Election Results 08-24*. https://github.com/tonmcg/US_County_Level_Election_Results_08-24 The copies committed in `./data-sources/` (3,152 and 3,160 rows). **Not authoritative, by its own account** — the README states the figures were scraped from news organizations' results pages. Every arrow on every national map in this chapter is drawn from it, on purpose; the section "The same election, as the states published it" says why it was kept rather than replaced.
- County presidential returns, 2020 and 2024, as published by the 51 state chief election offices: `pres2020_counties_official.csv` and `pres2024_counties_official.csv`, 3,197 rows each, with one row per jurisdiction and year in `./county-returns/data/provenance.csv` giving the URL, the file format, the row count and whether the state calls the figures certified. Used in this chapter as a ruler only: nothing is drawn from it. 50 of its 2024 rows carry no FIPS code, because the units they describe — District of Columbia wards, Rhode Island towns, Kansas City, Maine's statewide military and overseas ballots — have none, and the assembly rule forbids inventing one.
- Associated Press, certified state-level presidential totals for 2024, held as `crosscheck_ap_2024.csv` (51 rows). Used only to score the two county files against a third that neither was built from.
- County presidential returns, 2016 — MIT Election Data and Science Lab, *County Presidential Election Returns 2000-2016*, file version 20190722. <https://github.com/MEDSL/county-returns> Fetched 2026-08-12: 9,474 rows in the 2016 slice, 3,152 county codes after pivoting. Compiled from official state election records. This replaced the 2016 file from the McGovern repository, which is an election-night scrape missing 6,578,910 late-counted votes; the comparison is in `data/audit-2016-source.R` and `source_audit_2016.csv`.
- Referee for that comparison — Jason Timm, *PresElectionResults*, state-level returns. <https://github.com/jaytimmm/PresElectionResults> Used only to score the two 2016

candidates against a file derived from neither; it is the source behind this course's historical-campaigns data and is read from there.

- Federal Election Commission, *Election Results and Voting Information* — official federal returns, published by state and congressional district and **not by county**. <https://www.fec.gov/introduction-campaign-finance/election-results-and-voting-information/>

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`correction_2016.csv`, `county_centroids.csv`, `ga_outline.csv`, `office_gap.csv`, `precinct_join_ladder.csv`, `precinct_join_residual.csv`, `provenance_audit.csv`, `us_outline.csv`, `wind_ga.csv`, `wind_ga_counties.csv`, `wind_us.csv`, `wind_us_1620.csv`, `wind_us_1624.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `correction_2016.csv` has `swing_old` and `swing_new` after a data fix, with `delta`. Sort by `delta`. How many counties changed direction once a leading zero was restored?
- `precinct_join_ladder.csv` shows a join improving rung by rung, with `pct_of_2020` matched. Where does it stop improving? What is left over?
- `precinct_join_residual.csv` names the counties that never matched, with `ballots_unmatched`. Add them up. What share of a state's ballots would a naive map silently drop?
- `office_gap.csv` compares the presidential and Senate margin in the same precinct. Sort by `gap`. Where did voters split their ticket, and does an arrow on a map show that at all?

Method

- Grofman, Bernard, and Jonathan Cervas. 2024. "Statistical Fallacies in Claims about 'Massive and Widespread Fraud' in the 2020 Presidential Election: Examining Claims Based on Aggregate Election Results." *Statistics and Public Policy* 11(1): 2289529. <https://doi.org/10.1080/2330443X.2023.2289529> Section 2.1.2, "Failing to Weight Units," is why Figure 6 carries two fitted lines rather than one, and why every county share in this chapter is printed beside the corresponding vote share.
- The encoding constants — 8 degrees per point everywhere, full arrow length at 25,000 net votes on the national maps and 8,000 in Georgia, a 15% length floor and a 90-degree rotation cap — are declared once in `data/encoding.R` and consumed by every figure here. None is derived from the data it draws. `data/reencode.R` re-applies that file to the built CSVs, so the drawing rule can change without re-fetching a vote.

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original sources, without any starting files. Please do the following and show your work.

THE FIRST SOURCE. County-level presidential returns for 2020 and 2024 from Tony McGovern's compilation, served raw from GitHub:

https://raw.githubusercontent.com/tonmcg/US_County_Level_Election_Results_08-24/master/2020_US_County_Level_Presidential_Results.csv

and the matching 2024 file beside it. Each row is one county, keyed by FIPS code, with votes by candidate. No account or key is needed. Its own README says the numbers were scraped from news organizations and are not authoritative – there is no federal publisher of county presidential returns, and that is part of the story.

THE SECOND SOURCE. The Census Bureau's population-weighted county centroids, one plain text file:

https://www2.census.gov/geo/docs/reference/cenpop2020/county/CenPop2020_Mean_CO.txt

Each row is a county with the longitude and latitude of its people, not of its empty land. The header carries an invisible byte-order mark; read it as UTF-8-with-BOM or the first column name arrives mangled.

THE THIRD SOURCE. For Georgia alone, the state's own certified returns, exported as JSON by the Secretary of State:

<https://results.sos.ga.gov/cdn/results/Georgia/export-2024NovGen.json>

WHAT TO BUILD.

1. A national table of county vote swings, 2020 to 2024, joined to the centroids by FIPS. Do not trust the join: Alaska reports State House districts under invented FIPS codes, three of which collide with real borough codes; Connecticut swapped eight counties for nine planning regions between the files; and the code for the District of Columbia means the whole city in one file and a single ward in the other. A naive join succeeds everywhere and is wrong in all three places.

2. The same table for Georgia's 159 counties from the state's own export – the clean version, where every join key can be checked by eye.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- 3,221 county centroids in the census file
- 3,152 rows in the 2020 returns and 3,160 in 2024, of which only

3,104 describe the same place in both years

– 159 Georgia counties, joining both years with zero misses

The GitHub compilation takes corrections as counties certify and amend; small drifts in its row counts are revisions, not errors.